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|----------|------------|---|----------|-----|
| 3 | (a) | Resultant force (and resultant torque) is zero
Weight (down) = force from/due to spring (up) | B1
B1 | [2] |
| | (b) | (i) 0.2, 0.6, 1.0 s (<i>one of these</i>) | A1 | [1] |
| | | (ii) 0, 0.8 s (<i>one of these</i>) | A1 | [1] |
| | | (iii) 0.2, 0.6, 1.0 s (<i>one of these</i>) | A1 | [1] |

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- (c) (i) Hooke's law: extension is proportional to the force (*not mass*)
 Linear/straight line graph hence obeys Hooke's law
 B1
 B1 [2]
- (ii) Use of the gradient (*not just $F = kx$*)
 $K = (0.4 \times 9.8) / 15 \times 10^{-2}$
 $= 26(.1) \text{ N m}^{-1}$
 C1
 M1
 A0 [2]
- (iii) *either* energy = area to left of line *or* energy = $\frac{1}{2} ke^2$
 $= \frac{1}{2} \times [(0.4 \times 9.8) / 15 \times 10^{-2}] \times (15 \times 10^{-2})^2$
 $= 0.294 \text{ J}$ (*allow 2 s.f.*)
 C1
 C1
 A1 [3]

4 (a) (i) $R \equiv V^2 / P$ *or* $P \equiv IV$ and $V \equiv IR$ C1